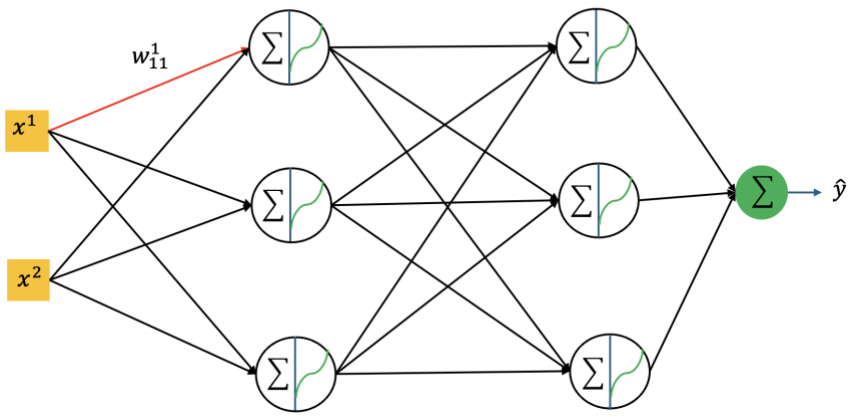
HW4 — CS5402: Introduction to Data Mining

Deadline: **04/10/2026**, midnight

Task 1: Theory questions [45%] :

According to the Neural Network structure shown in the following figure, and assume that:

1. All weights are set as: 0.5
2. Activation function is LeakyReLU: $f(x) = \begin{cases} 0.2 * x, & x < 0 \\ x, & x \geq 0 \end{cases}$
3. Loss function is defined as: $L = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2$, where y_i is the ground truth and $\hat{y}_i$ is the output of the Neural Network.



Given a dataset $D = \{(x_1, y_1), (x_2, y_2), (x_3, y_3), (x_4, y_4)\}$, and

- $x_1 = (1, 3)^T, y_1 = 5$
- $x_2 = (-1, 2)^T, y_2 = 3$
- $x_3 = (2, 6)^T, y_3 = 7$
- $x_4 = (4, 2)^T, y_4 = 2$

- Q1: [20%] [Forward Propagation] calculate the corresponding output $\hat{y}_1, \hat{y}_2, \hat{y}_3, \hat{y}_4$ [Show your intermediate calculation steps.
- Q2: [25%] [Backward Propagation] calculate the loss function and gradient: $\frac{\partial L}{\partial w_{11}^1}$ (red weight in the structure) [Show your intermediate calculation steps.]

Task 2: Custom Convolution Layer Implementation [55%]

In this assignment, you will implement a **custom 2D convolution layer from scratch** with the following constraints:

1. Support **arbitrary kernel sizes**
2. **No padding**
3. Use a **non-uniform stride pattern**:
    - Horizontal stride: [1, 2, 1, 2, ...]
    - Vertical stride: [1, 2, 1, 2, ...]

Important Notes

- Input: 2D matrix (grayscale image)
- Kernel: 2D matrix
- No padding → valid convolution only
- Stride is **not constant**
- Output size depends on stride pattern